

Medusa: Simple LLM Inference Acceleration with Multiple Decoding Heads

Abstract

Speculative decoding accelerates LLM inference by generating draft tokens with a separate model. However, training and serving a secondary model introduces engineering complexity and memory overhead. We present Medusa, an architecture-efficient alternative that attaches multiple lightweight decoding heads to the frozen backbone LLM. Medusa generates multiple draft tokens in parallel without an auxiliary model and verifies them using tree-based attention in a single forward pass.

1. Introduction

Modern LLM serving is bottlenecked by the arithmetic intensity of memory-bound autoregressive decoding. While speculative decoding mitigates this with a draft model, managing two separate model graphs complicates scheduling and consumes GPU VRAM. Medusa solves this by adding K small linear or 1-layer MLP heads on top of the base LLM's final hidden representation.

2. Architecture & Approach

Medusa comprises two main components:

1. Medusa Heads: K lightweight heads ($H_1, H_2, \dots, H_K$) attached to the final hidden state h_t . Head H_k predicts the token at position $t + k + 1$.
2. Tree-Based Attention: Instead of proposing a single linear sequence of candidate tokens, Medusa combines top predictions from multiple heads into a candidate tree.
3. Verification Mask: A customized 2D attention mask evaluates the entire candidate tree simultaneously in a single forward pass of the backbone LLM.

Training of Medusa heads requires minimal compute:

- The base LLM parameters are completely frozen.
- Only the Medusa heads are trained using standard cross-entropy loss on target domain text.
- Parameter overhead is under 2% of the base model size.

3. Experimental Results

We benchmark Medusa across Vicuna-7B, Vicuna-13B, and LLaMA-2-70B on MT-Bench and GSM8K.

- Speedup: Achieves 2.2x to 3.1x wall-clock speedup over standard autoregressive decoding.
- Acceptance Length: Average accepted tokens per step reaches 2.5 to 3.2 tokens per forward pass.
- Comparison with Speculative Decoding: Medusa matches or exceeds the throughput of speculative decoding with draft models, while requiring zero additional model loading or KV-cache synchronization overhead.

4. Limitations

- Non-zero loss of precision if combined with typical acceptance threshold approximations; exact sampling requires strict rejection filtering.
- Requires dedicated fine-tuning of the head weights for each new base model checkpoint.